Task 3: Heart Disease Prediction

Objective: Build a classification model to predict whether a person is at risk of heart disease based on health data.

Dataset: Heart Disease UCI Dataset (loaded directly from UCI ML Repository)

In [9]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import (accuracy_score, confusion_matrix,
                             classification_report, roc_curve, auc, roc_auc_score)
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings("ignore")

sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

Step 2: Load the Heart Disease Dataset

In [10]:
```
# Load Heart Disease UCI dataset directly from URL
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data"

col_names = ["age", "sex", "cp", "trestbps", "chol", "fbs", "restecg",
             "thalach", "exang", "oldpeak", "slope", "ca", "thal", "target"]

df = pd.read_csv(url, names=col_names, na_values="?")
```

```python
print(f"Dataset loaded! Shape: {df.shape}")
print(f"Columns: {list(df.columns)}")
print()
print("First 5 rows:")
df.head()
```

```
Dataset loaded! Shape: (303, 14)
Columns: ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'targ
et']

First 5 rows:
```

Out[10]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	2
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0

Step 3: Data Cleaning & Preprocessing

Column Descriptions:

- `age` : Age of the patient
- `sex` : 1 = Male, 0 = Female
- `cp` : Chest pain type (0–3)
- `trestbps` : Resting blood pressure
- `chol` : Serum cholesterol
- `fbs` : Fasting blood sugar > 120 mg/dl
- `restecg` : Resting ECG results
- `thalach` : Maximum heart rate achieved
- `exang` : Exercise induced angina

- `oldpeak` : ST depression
- `slope` : Slope of peak exercise ST segment
- `ca` : Number of major vessels
- `thal` : Thalassemia type
- `target` : 0 = No disease, 1–4 = Disease present

In [11]:
```
print("=== Missing Values ===")
print(df.isnull().sum())
print()

# Fill missing values with median
df["ca"].fillna(df["ca"].median(), inplace=True)
df["thal"].fillna(df["thal"].median(), inplace=True)

# Binarize target: 0 = no disease, 1 = disease
df["target"] = (df["target"] > 0).astype(int)

print("After cleaning:")
print(f"  Missing values: {df.isnull().sum().sum()}")
print(f"  Target distribution:")
print(df["target"].value_counts().to_string())
print()
print("Dataset info:")
df.info()
```

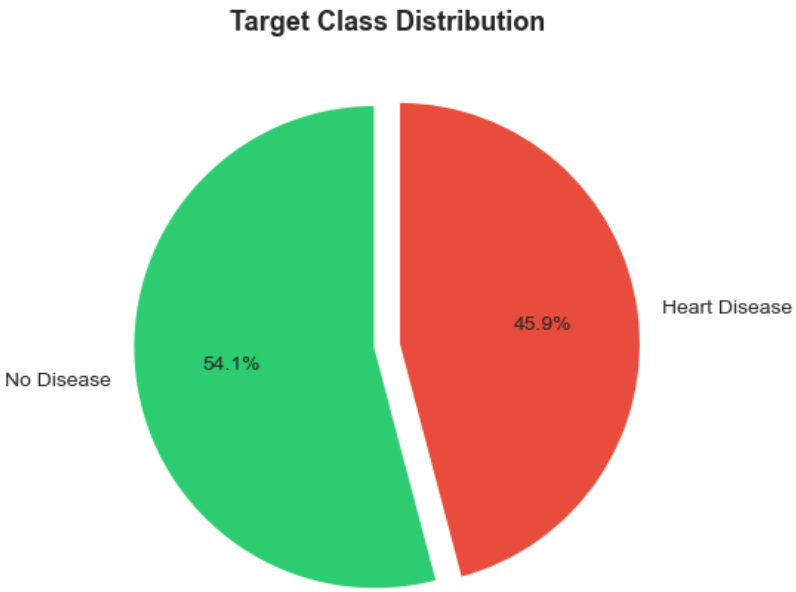
```
=== Missing Values ===
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          4
thal        2
target      0
dtype: int64

After cleaning:
  Missing values: 0
  Target distribution:
target
0    164
1    139

Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    float64
 1   sex       303 non-null    float64
 2   cp        303 non-null    float64
 3   trestbps  303 non-null    float64
 4   chol      303 non-null    float64
 5   fbs       303 non-null    float64
 6   restecg   303 non-null    float64
 7   thalach   303 non-null    float64
 8   exang     303 non-null    float64
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    float64
```

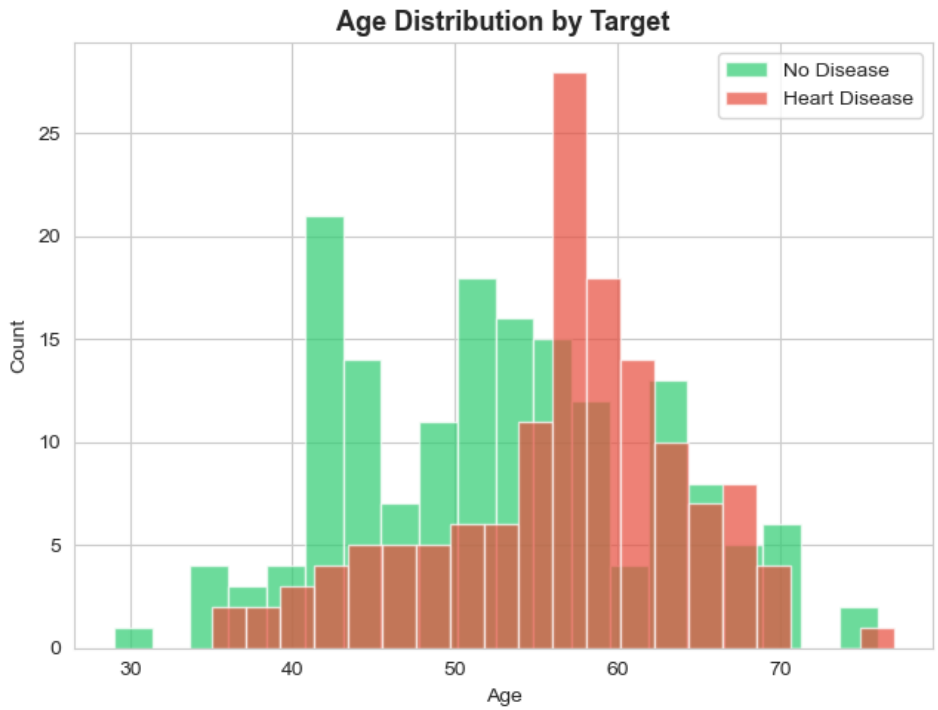
```
 11  ca        303 non-null    float64
 12  thal      303 non-null    float64
 13  target    303 non-null    int64
dtypes: float64(13), int64(1)
memory usage: 33.3 KB
```

Step 4: Exploratory Data Analysis (EDA)

In [12]:

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Target distribution
target_counts = df["target"].value_counts()
axes[0].pie(target_counts, labels=["No Disease", "Heart Disease"],
            autopct="%1.1f%%", colors=["#2ecc71", "#e74c3c"], startangle=90,
            explode=(0.05, 0.05))
axes[0].set_title("Target Class Distribution", fontsize=13, fontweight="bold")

# Age distribution by target
df[df["target"]==0]["age"].hist(ax=axes[1], bins=20, alpha=0.7,
                                color="#2ecc71", label="No Disease")
df[df["target"]==1]["age"].hist(ax=axes[1], bins=20, alpha=0.7,
                                color="#e74c3c", label="Heart Disease")
axes[1].set_title("Age Distribution by Target", fontsize=13, fontweight="bold")
axes[1].set_xlabel("Age")
axes[1].set_ylabel("Count")
axes[1].legend()

plt.tight_layout()
plt.savefig("eda_overview.png", dpi=100, bbox_inches="tight")
plt.show()
```

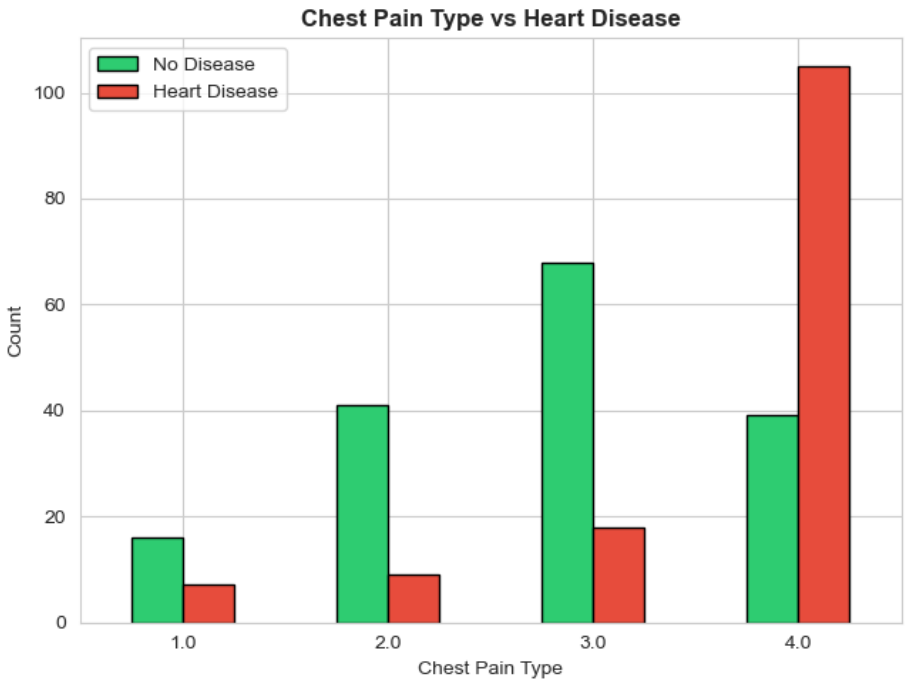


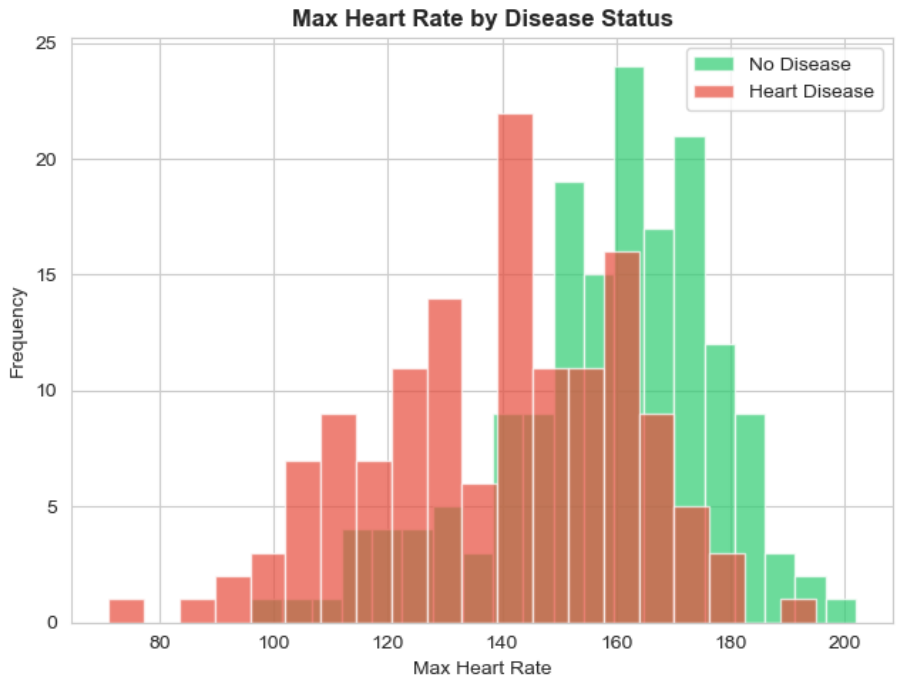


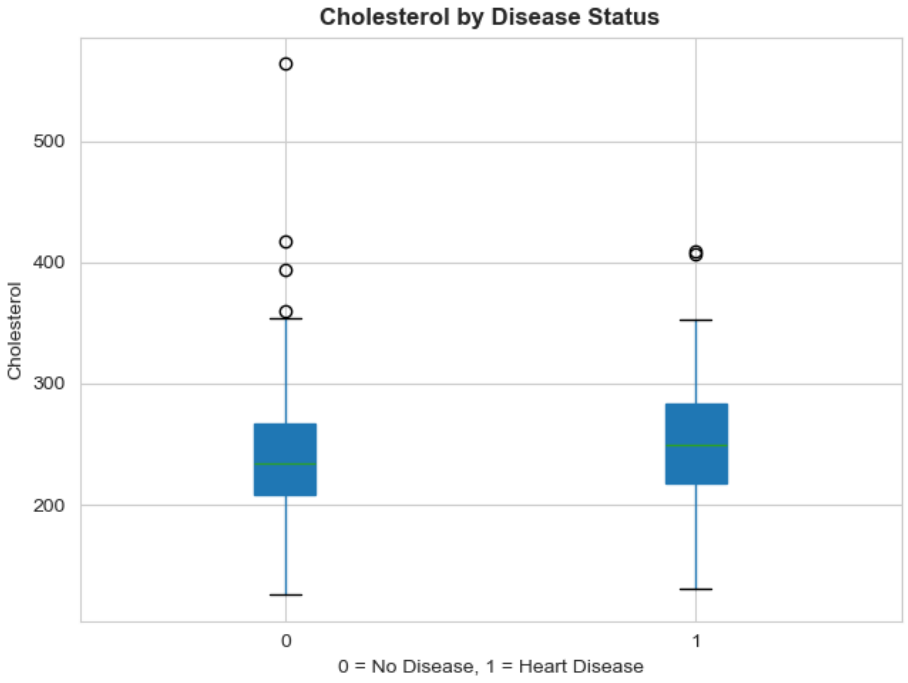
In [23]:
```
fig, axes = plt.subplots(2, 2, figsize=(13, 10))
axes = axes.flatten()


#  4. Chest Pain Type vs Target

cp_target = df.groupby(["cp", "target"]).size().unstack()

cp_target.plot(
    kind="bar",
    ax=axes[0],
    color=["#2ecc71", "#e74c3c"],
    edgecolor="black",
    rot=0
)

axes[0].set_title("Chest Pain Type vs Heart Disease", fontsize=12, fontweight="bold")
axes[0].set_xlabel("Chest Pain Type")
```

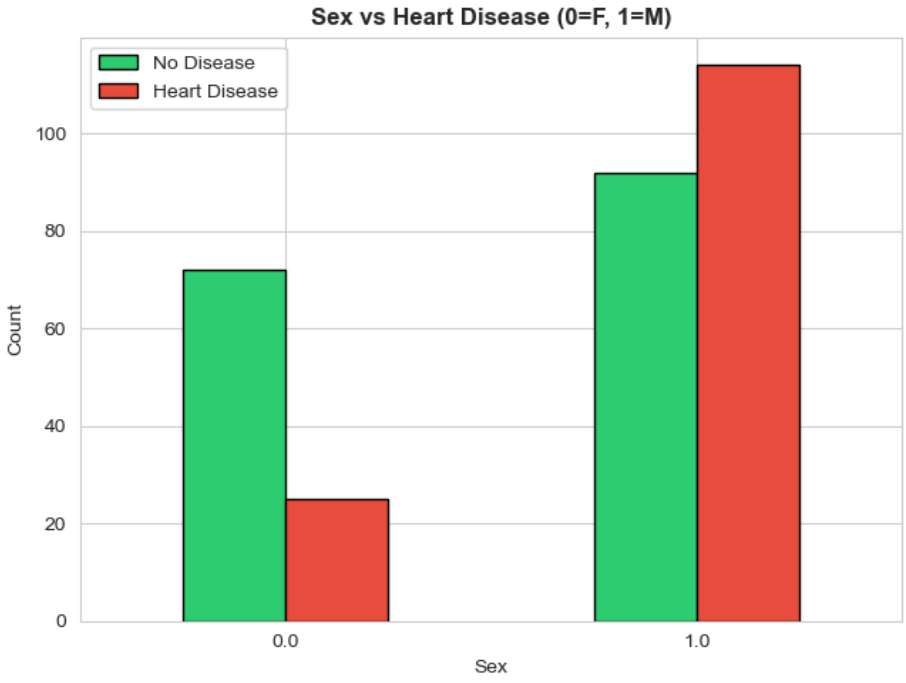
```python
axes[0].set_ylabel("Count")
axes[0].legend(["No Disease", "Heart Disease"])

#  5. Max Heart Rate Distribution

df[df["target"] == 0]["thalach"].hist(
    ax=axes[1],
    bins=20,
    alpha=0.7,
    color="#2ecc71",
    label="No Disease"
)

df[df["target"] == 1]["thalach"].hist(
    ax=axes[1],
    bins=20,
    alpha=0.7,
    color="#e74c3c",
    label="Heart Disease"
)

axes[1].set_title("Max Heart Rate by Disease Status", fontsize=12, fontweight="bold")
axes[1].set_xlabel("Max Heart Rate")
axes[1].set_ylabel("Frequency")
axes[1].legend()

#  6. Cholesterol Box Plot

df.boxplot(
    column="chol",
    by="target",
    ax=axes[2],
    patch_artist=True
)

axes[2].set_title("Cholesterol by Disease Status", fontsize=12, fontweight="bold")
axes[2].set_xlabel("0 = No Disease, 1 = Heart Disease")
axes[2].set_ylabel("Cholesterol")

# Remove automatic title from pandas
plt.suptitle("")
```

```python
# ==========================================
# 📌 7. Sex vs Target
# ==========================================
sex_target = df.groupby(["sex", "target"]).size().unstack()

sex_target.plot(
    kind="bar",
    ax=axes[3],
    color=["#2ecc71", "#e74c3c"],
    edgecolor="black",
    rot=0
)

axes[3].set_title("Sex vs Heart Disease (0=F, 1=M)", fontsize=12, fontweight="bold")
axes[3].set_xlabel("Sex")
axes[3].set_ylabel("Count")
axes[3].legend(["No Disease", "Heart Disease"])

# ==========================================
# 📌 8. Layout + Save
# ==========================================
plt.tight_layout()

plt.savefig(
    "eda_features.png",
    dpi=100,
    bbox_inches="tight"
)

plt.show()

print("✅ EDA Visualization Completed & Saved as 'eda_features.png'")
```


Chest Pain Type vs Heart Disease
No Disease
Heart Disease
Count
0
20
40
60
80
100
1.0
2.0
3.0
4.0
Chest Pain Type


Max Heart Rate by Disease Status
No Disease
Heart Disease
Frequency
0
5
10
15
20
25
80
100
120
140
160
180
200
Max Heart Rate

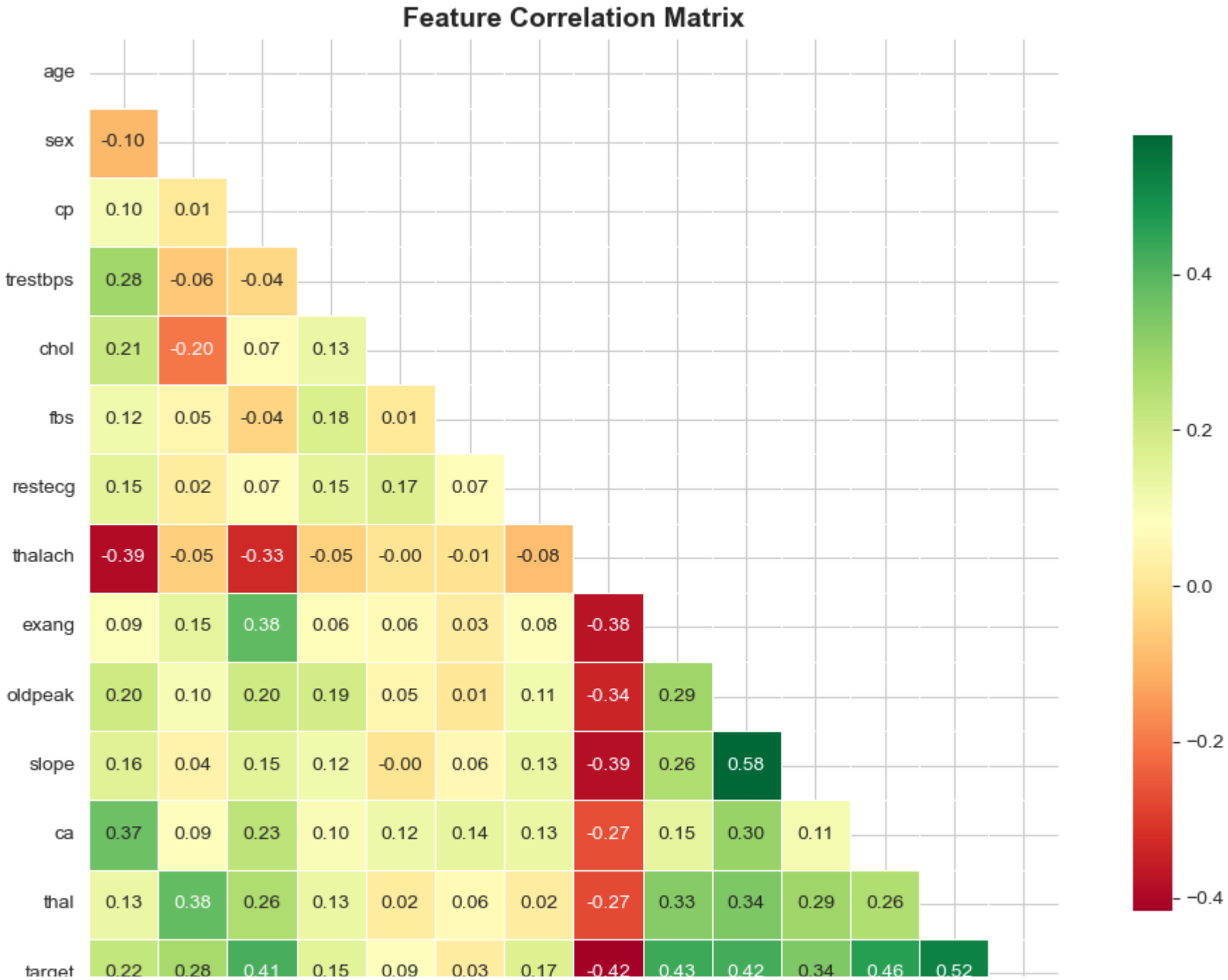

Cholesterol by Disease Status
Cholesterol
200
300
400
500
0
1
0 = No Disease, 1 = Heart Disease

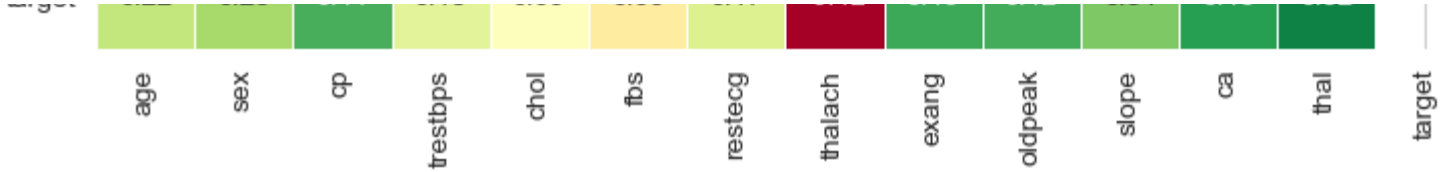

Sex vs Heart Disease (0=F, 1=M)
No Disease
Heart Disease
Count
0
20
40
60
80
100
0.0
1.0
Sex

```
✅ EDA Visualization Completed & Saved as 'eda_features.png'
```

Correlation Heatmap

In [14]:
```
plt.figure(figsize=(11, 8))
corr = df.corr()
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="RdYlGn",
            mask=mask, linewidths=0.5, square=True,
            cbar_kws={"shrink": 0.8})
plt.title("Feature Correlation Matrix", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("correlation_heatmap.png", dpi=100, bbox_inches="tight")
plt.show()
```


Feature Correlation Matrix
age
sex -0.10
cp 0.10 0.01
trestbps 0.28 -0.06 -0.04
chol 0.21 -0.20 0.07 0.13
fbs 0.12 0.05 -0.04 0.18 0.01
restecg 0.15 0.02 0.07 0.15 0.17 0.07
thalach -0.39 -0.05 -0.33 -0.05 -0.00 -0.01 -0.08
exang 0.09 0.15 0.38 0.06 0.06 0.03 0.08 -0.38
oldpeak 0.20 0.10 0.20 0.19 0.05 0.01 0.11 -0.34 0.29
slope 0.16 0.04 0.15 0.12 -0.00 0.06 0.13 -0.39 0.26 0.58
ca 0.37 0.09 0.23 0.10 0.12 0.14 0.13 -0.27 0.15 0.30 0.11
thal 0.13 0.38 0.26 0.13 0.02 0.06 0.02 -0.27 0.33 0.34 0.29 0.26
target 0.22 0.28 0.41 0.15 0.09 0.03 0.17 -0.42 0.43 0.42 0.34 0.46 0.52
0.4
0.2
0.0
−0.2
−0.4



Step 5: Feature Selection & Train/Test Split

In [15]:
```
feature_cols = ["age", "sex", "cp", "trestbps", "chol", "fbs", "restecg",
                "thalach", "exang", "oldpeak", "slope", "ca", "thal"]

X = df[feature_cols]
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled  = scaler.transform(X_test)

print(f"Training samples : {len(X_train)}")
print(f"Testing  samples : {len(X_test)}")
print(f"Train class balance: {dict(y_train.value_counts())}")
print(f"Test  class balance: {dict(y_test.value_counts())}")
```

```
Training samples : 242
Testing  samples : 61
Train class balance: {0: np.int64(131), 1: np.int64(111)}
Test  class balance: {0: np.int64(33), 1: np.int64(28)}
```

Step 6: Train Classification Models

In [16]:
```
# Model 1: Logistic Regression
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train_scaled, y_train)
lr_preds = lr.predict(X_test_scaled)
```

```
lr_proba = lr.predict_proba(X_test_scaled)[:, 1]

# Model 2: Decision Tree
dt = DecisionTreeClassifier(max_depth=5, random_state=42)
dt.fit(X_train_scaled, y_train)
dt_preds = dt.predict(X_test_scaled)
dt_proba = dt.predict_proba(X_test_scaled)[:, 1]

print("Models trained!")
print()
print("LOGISTIC REGRESSION:")
print(f"  Accuracy: {accuracy_score(y_test, lr_preds)*100:.2f}%")
print(f"  AUC-ROC:  {roc_auc_score(y_test, lr_proba):.4f}")

print()
print("DECISION TREE:")
print(f"  Accuracy: {accuracy_score(y_test, dt_preds)*100:.2f}%")
print(f"  AUC-ROC:  {roc_auc_score(y_test, dt_proba):.4f}")
```

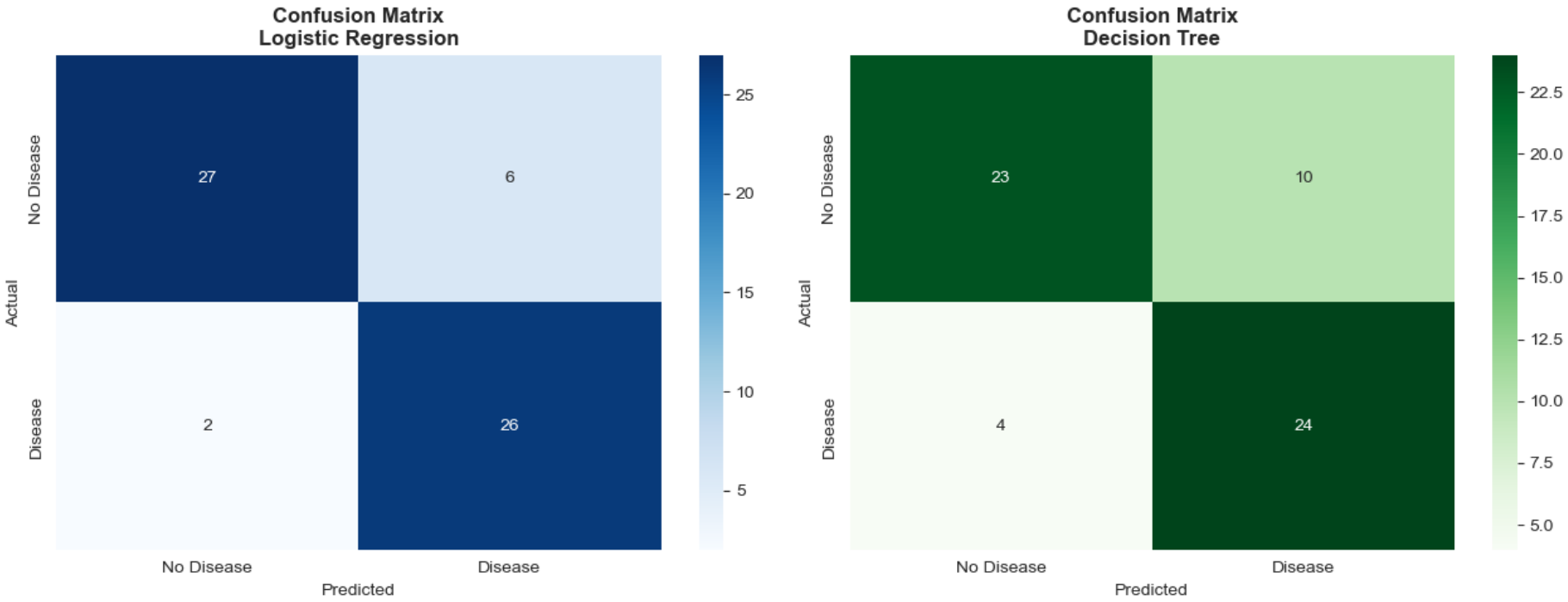
```
Models trained!

LOGISTIC REGRESSION:
  Accuracy: 86.89%
  AUC-ROC:  0.9513

DECISION TREE:
  Accuracy: 77.05%
  AUC-ROC:  0.8030
```

Step 7: Evaluation — Confusion Matrix, ROC Curve & Classification Report

In [17]:

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

for ax, preds, name, color in [
    (axes[0], lr_preds, "Logistic Regression", "Blues"),
    (axes[1], dt_preds, "Decision Tree",       "Greens")
]:
    cm = confusion_matrix(y_test, preds)
    sns.heatmap(cm, annot=True, fmt="d", cmap=color, ax=ax,
                xticklabels=["No Disease", "Disease"],
```

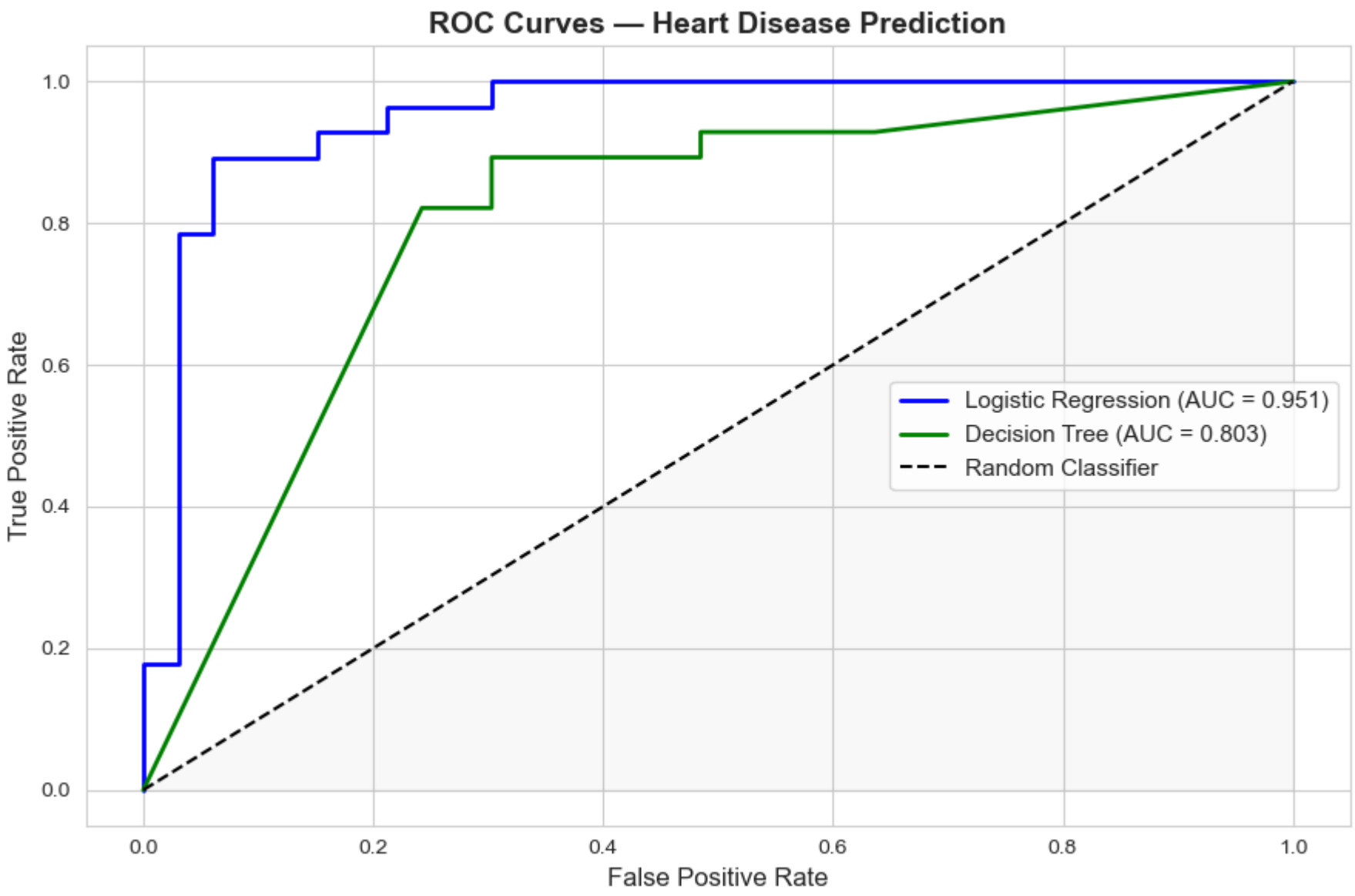
```
                yticklabels=["No Disease", "Disease"])
    ax.set_title(f"Confusion Matrix\n{name}", fontsize=12, fontweight="bold")
    ax.set_ylabel("Actual")
    ax.set_xlabel("Predicted")

plt.tight_layout()
plt.savefig("confusion_matrices.png", dpi=100, bbox_inches="tight")
plt.show()
```



In [18]:

```
# ROC Curves
plt.figure(figsize=(9, 6))

for proba, name, color in [
    (lr_proba, "Logistic Regression", "blue"),
    (dt_proba, "Decision Tree",       "green")
]:
    fpr, tpr, _ = roc_curve(y_test, proba)
    auc_score   = auc(fpr, tpr)
    plt.plot(fpr, tpr, color=color, linewidth=2,
```

```
             label=f"{name} (AUC = {auc_score:.3f})")

plt.plot([0,1], [0,1], "k--", linewidth=1.5, label="Random Classifier")
plt.fill_between([0,1], [0,1], alpha=0.05, color="gray")
plt.xlabel("False Positive Rate", fontsize=12)
plt.ylabel("True Positive Rate", fontsize=12)
plt.title("ROC Curves — Heart Disease Prediction", fontsize=14, fontweight="bold")
plt.legend(fontsize=11)
plt.tight_layout()
plt.savefig("roc_curves.png", dpi=100, bbox_inches="tight")
plt.show()
```



In [19]:
```
print("=== LOGISTIC REGRESSION REPORT ===")
print(classification_report(y_test, lr_preds,
      target_names=["No Disease", "Heart Disease"]))

print("=== DECISION TREE REPORT ===")
```

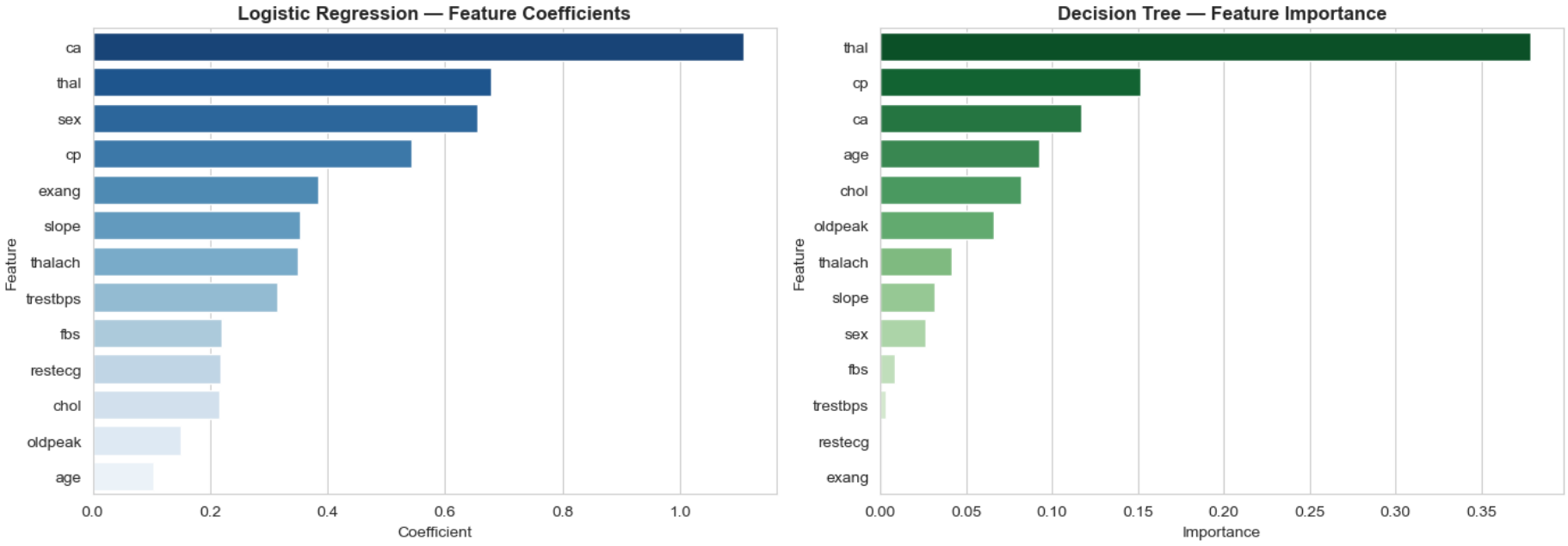
```python
print(classification_report(y_test, dt_preds,
      target_names=["No Disease", "Heart Disease"]))
```

```
=== LOGISTIC REGRESSION REPORT ===
               precision    recall  f1-score   support

   No Disease       0.93      0.82      0.87        33
Heart Disease       0.81      0.93      0.87        28

     accuracy                           0.87        61
    macro avg       0.87      0.87      0.87        61
 weighted avg       0.88      0.87      0.87        61

=== DECISION TREE REPORT ===
               precision    recall  f1-score   support

   No Disease       0.85      0.70      0.77        33
Heart Disease       0.71      0.86      0.77        28

     accuracy                           0.77        61
    macro avg       0.78      0.78      0.77        61
 weighted avg       0.78      0.77      0.77        61
```

Step 8: Feature Importance

In [20]:
```python
# Logistic Regression coefficients
lr_importance = pd.DataFrame({
    "Feature": feature_cols,
    "Coefficient": np.abs(lr.coef_[0])
}).sort_values("Coefficient", ascending=False)

# Decision Tree feature importances
dt_importance = pd.DataFrame({
    "Feature": feature_cols,
    "Importance": dt.feature_importances_
}).sort_values("Importance", ascending=False)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
```

```
sns.barplot(data=lr_importance, x="Coefficient", y="Feature",
            palette="Blues_r", ax=axes[0])
axes[0].set_title("Logistic Regression — Feature Coefficients",
                  fontsize=12, fontweight="bold")

sns.barplot(data=dt_importance, x="Importance", y="Feature",
            palette="Greens_r", ax=axes[1])
axes[1].set_title("Decision Tree — Feature Importance",
                  fontsize=12, fontweight="bold")

plt.tight_layout()
plt.savefig("feature_importance_heart.png", dpi=100, bbox_inches="tight")
plt.show()

print("Top 5 features (Decision Tree):")
print(dt_importance.head(5).to_string(index=False))
```



```
Top 5 features (Decision Tree):
Feature  Importance
   thal    0.378859
     cp    0.151595
     ca    0.116874
    age    0.092980
   chol    0.082157
```

Step 9: Key Insights & Conclusions

In [21]:
```
lr_acc = accuracy_score(y_test, lr_preds)*100
dt_acc = accuracy_score(y_test, dt_preds)*100
lr_auc = roc_auc_score(y_test, lr_proba)
dt_auc = roc_auc_score(y_test, dt_proba)

print("="*55)
print("KEY INSIGHTS - HEART DISEASE PREDICTION")
print("="*55)
print()
print(f"Dataset: UCI Heart Disease | {len(df)} patients")
print()
print("MODEL RESULTS:")
print(f"  Logistic Regression -> Accuracy: {lr_acc:.1f}% | AUC: {lr_auc:.3f}")
print(f"  Decision Tree       -> Accuracy: {dt_acc:.1f}% | AUC: {dt_auc:.3f}")
print()
print("KEY CLINICAL FINDINGS:")
print("  - Chest pain type (cp) is the strongest predictor")
print("  - Max heart rate (thalach) inversely linked to disease")
print("  - Males more frequently diagnosed with heart disease")
print("  - oldpeak (ST depression) is a significant risk indicator")
print()
print("CONCLUSION:")
print("  Logistic Regression is interpretable & performs well.")
print("  These features can support early risk screening.")
```

```
=========================================================
KEY INSIGHTS - HEART DISEASE PREDICTION
=========================================================

Dataset: UCI Heart Disease | 303 patients

MODEL RESULTS:
  Logistic Regression -> Accuracy: 86.9% | AUC: 0.951
  Decision Tree       -> Accuracy: 77.0% | AUC: 0.803

KEY CLINICAL FINDINGS:
  - Chest pain type (cp) is the strongest predictor
  - Max heart rate (thalach) inversely linked to disease
  - Males more frequently diagnosed with heart disease
  - oldpeak (ST depression) is a significant risk indicator

CONCLUSION:
  Logistic Regression is interpretable & performs well.
  These features can support early risk screening.
```

In [ ]: